

Problem-Solution Fit

Date	26 June 2026
Team ID	PNT2026TMID-EG01
Project Name	EduGenie - AI-Powered Educational Assistant
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Students (school, college, university), self-learners, and educators across all academic levels and subject domains seeking AI-powered educational support.	6. CUSTOMER LIMITATIONS [CL] EG. BUDGET, DEVICES Limited data plans or slow internet; personal devices (mobile/laptop) with varying specs; free-tier users with restricted AI API quotas; non-technical users unfamiliar with AI tools.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] ChatGPT: Pro – general Q&A; Con – not education-focused, no structured roadmap. Google Search: Pro – broad info; Con – unfiltered, no personalization. Khan Academy: Pro – structured; Con – no AI Q&A or custom quiz generation.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Students struggle to get quick, accurate answers on academic topics (daily). Concept explanations are hard to find at the right difficulty level (frequent). No personalized learning roadmap available (weekly). Self-assessment tools are generic and not topic-specific (weekly).	9. PROBLEM ROOT / CAUSE [RC] Existing tools are either too generic (search engines) or too rigid (static course platforms). There is no lightweight, AI-powered assistant that combines Q&A, explanation, quiz generation, and roadmap creation in one unified interface.	7. BEHAVIOR + ITS INTENSITY [BE] Students frequently search for concept explanations and answers online (high intensity – multiple times daily). Self-testing via quizzes is done occasionally (medium). Planning a structured learning path is rare due to lack of easy tools (low frequency, high need).
3. TRIGGERS TO ACT [TR] Upcoming exams or assignments requiring quick concept clarity. Inability to understand a topic after reading textbooks. Desire to self-assess understanding before a test. Starting to learn a new subject with no clear path.	10. YOUR SOLUTION [SL] EduGenie – an AI-powered educational assistant built on FastAPI and Gemini 1.5 Pro that provides intelligent Q&A, concept explanation at multiple difficulty levels, topic-specific quiz generation, text summarization, and personalized learning roadmap recommendations through a simple, interactive web interface.	8. CHANNELS OF BEHAVIOR [CH] ONLINE: Web browser (desktop/mobile) via the EduGenie web app; accessed at 127.0.0.1:8000 locally or via hosted server.
4. EMOTIONS BEFORE / AFTER [EM] BEFORE: Frustrated, overwhelmed, anxious about not understanding topics, lack of direction in learning. AFTER: Confident, focused, motivated with a clear learning path, instant answers, and verified self-assessment.		OFFLINE: Word-of-mouth among students; classroom or study group recommendations; shared learning roadmaps via screenshots.